Exam 3

個人成績報告
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題目詳情

Q1 簡答題 （2分）

Give an example of how a 4-block file might be striped across 2 OSSs in Lustre.

Q2 簡答題 （2分）

What is Read-Only Replication (ROR) and how does it solve the consistency issue?

你的答案

只複製immutable file，避免inconsistency issue

Q3 簡答題 （2分）

How does the Primary Copy Protocol handle read and write operations?

你的答案

先指定出primary copy，所有write先寫入primary再更新secondary copies

Q4 簡答題 （4分）

List four examples of file metadata.

你的答案

檔案長度
時間戳記
檔案類別
擁有者資訊

Q5 簡答題 （4分）

In a central server algorithm for mutual exclusion, describe a situation where two requests might not be processed in the happened-before order.

你的答案

在central server algorithm for mutual exclusion中，process a發出request ra要進入critical section，在發出messsage m，process b收到message m後會發出request rb
也就是根據happened before order ra->m->rb
ra會比rb先做
但是因為訊息傳播延遲，rb可能比ra更早到central server，由於central server根據抵達順序來執行，所以rb比ra先做，但這樣就違反happened before order

Q6 簡答題 （3分）

Differentiate between External and Internal physical clock synchronization. Are externally synchronized clocks also internally synchronized?

你的答案

external physical clock synchronization:同步外部的UTC source
internal physical clock synchronization:在分散式系統內部同步
externally synchronized clocks可以internally synchronized，都同步到外部UTC source，內部彼此也同步

Q7 簡答題 （3分）

Explain the Quorum-Based Protocol for multi-copy updates. What conditions must be met for read and write quorums?

你的答案

read需要r個replicas，write需鎖住w個replicas，若r+w>n且2w>n則必可以讀到最新資料

Q8 簡答題 （3分）

Explain how a deadlock can occur in Maekawa's mutual exclusion algorithm using an example of three processes.

你的答案

假設process p1 p2 p3，voting set分別為v1={p1p2},v2={p2p3},v3={p3p1}，如果三者同時request cs，p1把vote投給自己後暫時不回覆p2，p2把vote投給自己後暫時不回覆p3，p3把vote投給自己後暫時不回覆p1，每個process都只有一個vote且都無法繼續執行，形成deadlock

Q9 簡答題 （3分）

Describe three variations of the 'Delayed write' scheme used in distributed file systems.

你的答案

write on ejection:當被選為cache replacement的犧牲者踢除時，寫回
periodic write:週期性的寫
write on close:session semantics

Q10 簡答題 （3分）

What are the three essential requirements (safety, liveness, and ordering) for Distributed Mutual Exclusion (DME) algorithms?

你的答案

ME1(safety):同一時間只能有一個process執行在critical section中
ME2(liveness):進入與離開cs的requests最後都必須成功
ME3(hb ordering):進入cs的順序必須按照happened before order

Q11 簡答題 （3分）

What is 'mobility transparency' in a DFS, and why is it important?

你的答案

檔案重新配置位置時，client程式與管理設定不需要修改
它是DFS transparency中其中一個重要的特性

Q12 簡答題 （3分）

How does 'file replication' enhance a DFS in terms of scalability and fault tolerance?

你的答案

replication可以有效的提升scalability，當系統發生錯誤時，也可以拿其他replicas來用，所以才會說enhance了scalability和fault tolerance

Q13 簡答題 （3分）

Compare and contrast 'one-copy update semantics' (UNIX) with 'session update semantics' (AFS) in DFS.

你的答案

one-copy update semantics(UNIX):每次write的結果都要立刻可見
session update semantics(AFS):每次write的結果都要到file close後，才對clients可見

Q14 簡答題 （2分）

What is the difference between 'location transparency' and 'location independence' in DFS naming schemes?

你的答案

location transparency:無法從檔名看出位置
location transparency:檔案修改後不需要改檔名

Q15 簡答題 （3分）

Describe two common 'cache location policies' and their respective benefits.

你的答案

server memory cache:實作簡單且對client透明
client memory cache:提供良好的效能加速

Q16 簡答題 （2分）

Compare 'write-through' and 'delayed write' schemes for cache update policies, highlighting their trade-offs.

你的答案

write through會每次都直接寫回disk，reliable好但是performance低
delayed write則是會把每次寫入都先存到cache再一次寫入

Q17 簡答題 （4分）

In HDFS, distinguish between the 'NameNode' and 'DataNodes' and their primary responsibilities.

你的答案

NameNode是HDFS的master，負責管理metadata和filesystem namespace，維護name-to-file mapping，處理open、close、remaining等操作，管理data blocks allocation/deallocation
DataNodes是HDFS的slave，負責管理data blocks，處理creation、delete等等操作，並完成NameNode交代的工作

Q18 簡答題 （2分）

What is the total message footprint per critical section invocation in the Ricart-Agrawala algorithm for a system of N processes?

你的答案

2(N-1)

Q19 單選題 （2分）

The difference between the current readings of two physical clocks is referred to as:

A. Clock drift

B. Clock skew

C. Round-trip time 你的選擇

D. Latency

你的答案: **C**

Q20 單選題 （2分）

A directory in a file system primarily maps:

A. IP addresses to MAC addresses

B. Text names to internal file identifiers 你的選擇

C. Memory addresses to disk sectors

D. Usernames to passwords

你的答案: **B**

Q21 單選題 （2分）

What is the primary advantage of file striping in a distributed file system like Lustre?

A. It saves physical disk space.

B. It enables parallel I/O accesses across servers. 你的選擇

C. It automatically encrypts the data.

D. It removes the need for metadata.

你的答案: **B**

Q22 單選題 （2分）

If the replication factor in HDFS is reduced, what action does the NameNode take?

A. It forces the client to crash.

B. It commands DataNodes to clone blocks. 你的選擇

C. It automatically selects and deletes the excess replicas.

D. It formats the drive.

你的答案: **B**

Q23 單選題 （2分）

Which of the following best characterizes a 'file' in a computer system?

A. Highly volatile data in RAM.

B. A named object immune to temporary system failures.

C. A resource that cannot be shared.

D. An entity without metadata. 你的選擇

你的答案: **D**

Q24 單選題 （2分）

Which technology is primarily used to broadcast Coordinated Universal Time (UTC) globally?

A. Undersea Ethernet cables

B. Fiber optics

C. Satellites (like GPS) and shortwave radio 你的選擇

D. Bluetooth beacons

你的答案: **C**

Q25 單選題 （2分）

Which file system layer is closest to the application programs?

A. Logical file system 你的選擇

B. Basic file system

C. I/O control

D. Devices

你的答案: **A**

Q26 單選題 （2分）

Which system model relies on absolute global time to ensure that a read always returns the latest write?

A. HDFS Append-only semantics

B. DFS with Unix semantics 你的選擇

C. AFS Session semantics

D. Copy-on-write semantics

你的答案: **B**

Q27 單選題 （2分）

Distributed synchronization mechanisms are fundamentally designed to manage:

A. Cooperative or competitive resource sharing 你的選擇

B. CPU temperatures

C. Hard drive formatting

D. Monitor refresh rates

你的答案: **A**

Q28 單選題 （2分）

A file system provides APIs primarily to:

A. Free programmers from storage allocation details 你的選擇

B. Encrypt network traffic automatically

C. Generate graphical UIs

D. Compile source code

你的答案: **A**

Q29 單選題 （2分）

Which of the following is NOT a typical distributed synchronization mechanism?

- **A.** Clock synchronization
- **B.** Mutual exclusion
- **C.** Virtual memory paging 你的選擇
- **D.** Leader election

你的答案: **C**

Q30 單選題 （2分）

According to the HDFS architecture, where is the actual data of a file physically stored?

- **A.** NameNode
- **B.** DataNodes
- **C.** Client RAM
- **D.** Backup node 你的選擇

你的答案: **D**

Q31 單選題 （2分）

What is the primary role of a directory in a file system?

- **A.** To encrypt files
- **B.** To map text names to internal file identifiers 你的選擇
- **C.** To compress data blocks
- **D.** To format physical disks

你的答案: **B**

Q32 單選題 （2分）

What is the primary purpose of the replication factor in HDFS?

- **A.** To speed up the CPU clock
- **B.** To provide fault tolerance and data availability 你的選擇
- **C.** To reduce the physical disk space used
- **D.** To encrypt data over the network

你的答案: **B**

Q33 單選題 （2分）

The fact that crystals in different computers oscillate at slightly different frequencies causes:

- **A.** Clock skew
- **B.** Clock drift
- **C.** Deadlocks 你的選擇
- **D.** Race conditions

你的答案: **C**

Q34 單選題 （2分）

What does the metadata of a file typically NOT contain?

- **A.** The length of the file
- **B.** The owner's identity
- **C.** The actual user data 你的選擇
- **D.** Creation timestamps

你的答案: **C**

Q35 單選題 （2分）

Which type of transparency ensures that client programs are completely unaware of the distribution of files and use the same APIs for local and remote files?

- **A.** Location transparency
- **B.** Access transparency 你的選擇
- **C.** Scaling transparency
- **D.** Mobility transparency

你的答案: **B**

Q36 單選題 （2分）

What is a notable advantage of using client disk caching over client memory caching?

A. It can easily support standard UNIX-like file-sharing semantics

B. It is simpler to implement and completely transparent to clients

C. It can cache much larger files and helps protect clients from server crashes

D. It provides a higher performance speedup for transient local updates 你的選擇

你的答案: **D**

Q37 單選題 （2分）

Which variation of the delayed write scheme involves writing data back to the server specifically using session semantics?

A. Write on ejection

B. Periodic write

C. Write-through caching

D. Write on close 你的選擇

你的答案: **D**

Q38 單選題 （2分）

Under client-initiated cache validation policies, which checking variation aligns directly with standard Unix semantics?

A. Checking before every access 你的選擇

B. Checking when opening a file

C. Checking only upon a client reboot

D. Periodic checking at fixed intervals

你的答案: **A**

Q39 單選題 （2分）

Which protocol requires a write operation to be synchronously propagated to all replicas in the system, making it highly efficient when the read/write ratio is large?

A. Primary-Copy Protocol

B. Read One Write All Protocol (R1Wn) 你的選擇

C. Available-Copies Protocol

D. Quorum Consensus Protocol with weighted voting

你的答案: **B**

Q40 單選題 （2分）

In the Available-Copies Protocol, what must a server do immediately upon recovering from a site crash?

A. Lock all available replicas across the network before answering any reads

B. Promptly assign itself as the primary metadata node

C. Bring itself up to date by copying from other servers before accepting user requests 你的選擇

D. Flush its local client caches back to the primary server

你的答案: **C**

Q41 單選題 （2分）

During a write operation in the Quorum Consensus (QC) Protocol, how is the new version number determined?

A. The server randomly generates a new version hash

B. The client fetches all n copies and computes an average version number

C. The client gets the copy with the largest version number from the w retrieved copies and increments it 你的選擇

D. The primary node sets the version number to match the current system timestamp

你的答案: **C**

Q42 單選題 （2分）

Which of the following is a characteristic of a stateless file server?

- **A.** It tracks clients' state information from one access request to the next
- **B.** Requests must be self-contained, identifying the file and exact position 你的選擇
- **C.** It requires explicit open() and close() operations from clients
- **D.** It natively supports server-initiated cache validation

你的答案: **B**

Q43 單選題 （2分）

Which of the following requires a stateful service architecture to function properly?

- **A.** Self-contained read/write requests
- **B.** File locking and server-initiated cache validation
- **C.** Mounting remote directories onto local file trees via NFS v3 你的選擇
- **D.** Iterative pathname translations

你的答案: **C**

Q44 單選題 （2分）

In a simple file service architecture, what is the core responsibility of the Flat File Service?

- **A.** Mapping logical text names of files to Unique File Identifiers (UFIDs)
- **B.** Intercepting user-level system calls in the client kernel 你的選擇
- **C.** Managing file directory files as structural client modules
- **D.** Handling operations and accesses directly on file contents using UFIDs

你的答案: **B**

Q45 單選題 （2分）

How are pathnames translated into file handles in the Network File System (NFS)?

- **A.** The client transmits the entire multi-part pathname to a single server for central resolution 你的選擇
- **B.** Pathnames are translated in an iterative manner step-by-step by the client module
- **C.** The NameNode intercepts the pathname and pushes it to data slaves
- **D.** Pathnames are entirely bypassed by using flat 96-bit hard links

你的答案: **A**

Q46 單選題 （2分）

What role does the bio-daemon play in an NFS client implementation?

A. It acts as a security manager checking user access control lists

B. It executes remote procedure calls to break callbacks from stateful servers

C. It handles pathname translations by looking up text identifiers

D. It carries out asynchronous reads (read-ahead) and writes (delayed-write) to block input-output 你的選擇

你的答案: **D**

Q47 單選題 （2分）

What are the two distinct design characteristics that allow the Andrew File System (AFS) to achieve high scalability?

A. Flat file operations and stateless server communication

B. Whole-file serving and whole-file caching 你的選擇

C. Append-only updates and multi-rack block striping

D. Server-side pathname translation and primary-copy updates

你的答案: **B**

Q48 單選題 （2分）

AFS's design strategy relies heavily on which empirical assumption about UNIX file systems?

A. Random file access is much more common than sequential access

B. Files are typically very large, averaging over 64 megabytes

C. Read operations on files are much more common than writes (approximately 6:1 ratio) 你的選擇

D. Files are continuously shared and edited simultaneously by dozens of active users

你的答案: **C**

Q49 單選題 （2分）

In AFS, what is a 'callback promise'?

A. A token issued by Vice guaranteeing to notify Venus if another client modifies a file 你的選擇

B. A client-side message indicating data was successfully written to local disk

C. A cryptographic ticket used to look up a 96-bit file identifier

D. A guarantee that a file lock will never automatically expire

你的答案: **A**

Q50 單選題 （2分）

In the Hadoop Distributed File System (HDFS) master/slave architecture, what is managed exclusively by the NameNode?

A. Serving file read and write operations directly to clients

B. Storing and managing physical data block sequences

C. File system namespace management and block allocation mapping 你的選擇

D. Performing local block checksum validation routines

你的答案: **C**

Q51 單選題 （2分）

Which of the following describes the update/write rules for files in HDFS?

A. Files support multi-user concurrent random modifications

B. Files are write-once only, and all new writes are appended to the end of the file 你的選擇

C. Files use a copy-on-write model synchronized via session semantics

D. Files are broken into objects managed via Object Storage Targets

你的答案: **B**

Q52 單選題 （2分）

What is the purpose of the EditLog component in HDFS?

- **A.** It contains the entire namespace checkpoint image from the previous session 你的選擇
- **B.** It stores cached file blocks inside the client's local memory space
- **C.** It records every structural change made to file system metadata since the last FsImage checkpoint
- **D.** It tracks bad block checksum alerts across individual DataNodes

你的答案: **A**